left, $A$-module) on the category of left $A$-modules are, in general, not exact (they are only exact on one side). The job of the functors $\mathrm{Tor}_i^A(M,?)$ and $\mathrm{Ext}_A^i(M,?)$ is to quantify the extent to which the functors $M\otimes_A ?$ and $\mathrm{Hom}_A(M,?)$ fail to be exact. Namely, these functors are defined as follows.

**Definition 8.2.1.** A **projective resolution** of $M$ is an exact sequence

$$\cdots \to P_2 \to P_1 \to P_0 \to M \to 0$$

such that all modules $P_i$, $i \geq 0$, are projective.

**Exercise 8.2.2.** Show that any module has a projective resolution (for example, one consisting of free modules).

**Definition 8.2.3.** Let $M$ be a right $A$-module, $P_\bullet$ a projective resolution of $M$, and $N$ a left $A$-module. For $i \geq 0$ we define $\mathrm{Tor}_i^A(M,N) = \mathrm{Tor}_i(M,N)$ to be the $i$th cohomology of the complex

$$\cdots \to P_2 \otimes_A N \to P_1 \otimes_A N \to P_0 \otimes_A N \to 0$$

induced by the resolution $P_\bullet$.

**Definition 8.2.4.** Let $M$ be a left $A$-module, $P_\bullet$ a projective resolution of $M$, and $N$ a left $A$-module. For $i \geq 0$ we define $\mathrm{Ext}_A^i(M,N) = \mathrm{Ext}^i(M,N)$ to be the $i$th cohomology of the complex

$$0 \to \mathrm{Hom}_A(P_0,N) \to \mathrm{Hom}_A(P_1,N) \to \mathrm{Hom}_A(P_2,N) \to \dots$$

induced by the resolution $P_\bullet$.

**Problem 8.2.5.** In this problem we will show that the cohomology groups $\mathrm{Tor}_i$ and $\mathrm{Ext}^i$ don't really depend on the projective resolution $P_\bullet$, in a fairly strong sense, which justifies the fact that we don't mention $P_\bullet$ in the notation for them.

Let $P_\bullet$, $Q_\bullet$ be two projective resolutions of $M$. Let $d_i^P : P_i \to P_{i-1}$, $d_i^Q : Q_i \to Q_{i-1}$ be the corresponding differentials; in particular, $d_0^P : P_0 \to M$, $d_0^Q : Q_0 \to M$.

(i) Show that there exists a homomorphism $f_0 : P_0 \to Q_0$ such that $d_0^Q \circ f_0 = d_0^P$.

(ii) Proceed to show by induction in $j$ that there exists a homomorphism $f_j : P_j \to Q_j$ such that $d_j^Q \circ f_j = f_{j-1} \circ d_j^P$.